

LAGUNA STATE POLYTECHNIC UNIVERSITY

Installation Guide

Enterprise Server Deployment and Operating System Installation

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BSIT 4-E

WEEK 3 – ITEP 414

Mr. Penaredondo

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1. Project Overview and Objectives

This project documents the deployment of a new Ubuntu Server virtual machine for ABC Startup Solutions. The server is intended as a reproducible foundation for file sharing, remote administration, database hosting, web hosting, and internal services.

Objectives

- Install Ubuntu Server LTS using the assigned virtual hardware.
- Configure hostname, a non-root administrative account, storage, networking, and OpenSSH.
- Verify login, hostname, IP addressing, connectivity, updates, and SSH service health.
- Compare BIOS and UEFI, explain Ubuntu startup, install Windows Server Evaluation, and compare enterprise server operating systems.

2. Requirements and VM Specifications

Required software: Oracle VirtualBox or VMware Workstation, Ubuntu Server LTS ISO, Windows Server Evaluation ISO, and internet access for updates.

Component	Configured value
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 virtual processors
Storage	40 GB VDI/VMDK
Network	NAT or instructor-approved bridged
Optical drive	Ubuntu Server LTS ISO

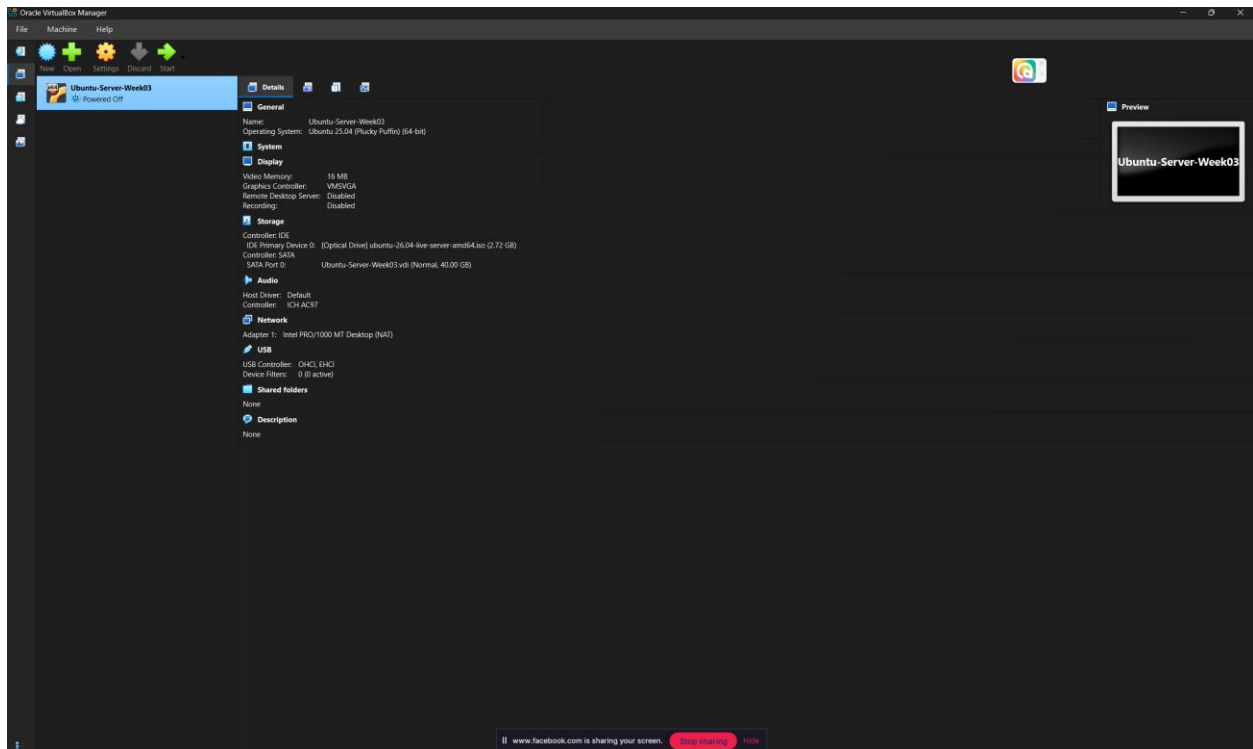
Pre-installation checklist

- Verify the ISO integrity when a published checksum is available.
- Confirm virtualization is enabled in the host firmware.
- Back up any existing data before repartitioning.
- Prepare a strong unique password and do not place it in this report.

3. Ubuntu Server Installation

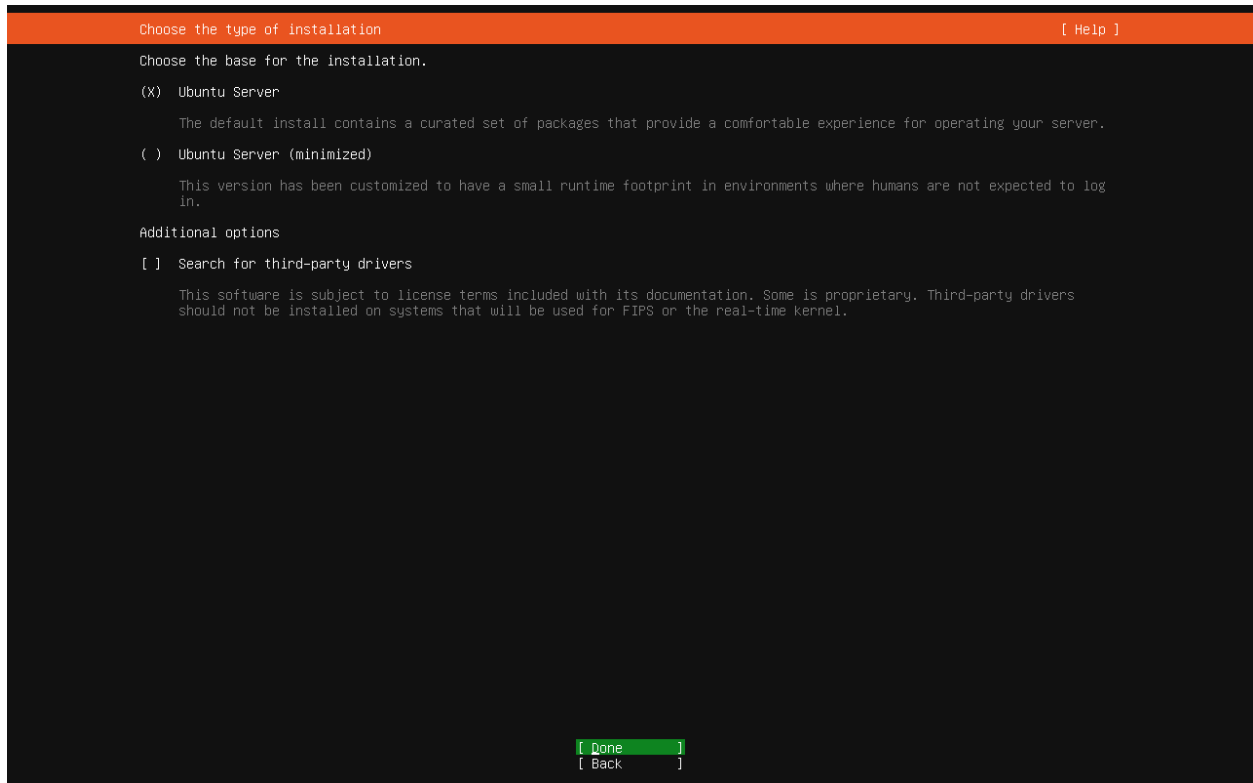
1. Create the virtual machine

Create Ubuntu-Server-Week03 in VirtualBox or VMware. Assign 4 GB RAM, 2 virtual processors, a 40 GB virtual disk, NAT networking, and attach the Ubuntu Server LTS ISO.



2. Start the installer

Boot from the ISO. Select English and the appropriate keyboard layout, such as English (US).



3. Configure storage

Select Guided - use an entire disk. Confirm that the selected device is the 40 GB virtual disk. Record the partition scheme, file system, and disk size.

Guided storage configuration

[Help]

Configure a guided storage layout, or create a custom one:

(X) Use an entire disk

[VBOX_HARDDISK_VBbba38e6b-78431c46 local disk 40.000G ▼]

[X] Set up this disk as an LVM group

[] Encrypt the LVM group with LUKS

Passphrase:

Confirm passphrase:

[] Also create a recovery key

The key will be stored as ~/recovery-key.txt in the live system and will be copied to /var/log/installer/ in the target system.

() Custom storage layout

[Done]

[Back]

4. Create the server identity

Set hostname to server01. Enter your full name, choose a non-root username, and use a strong unique password.

Profile configuration

[Help]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a password is still needed for sudo.

Your name:

Your servers name:
The name it uses when it talks to other computers.

Pick a username:

Choose a password:

Confirm your password:

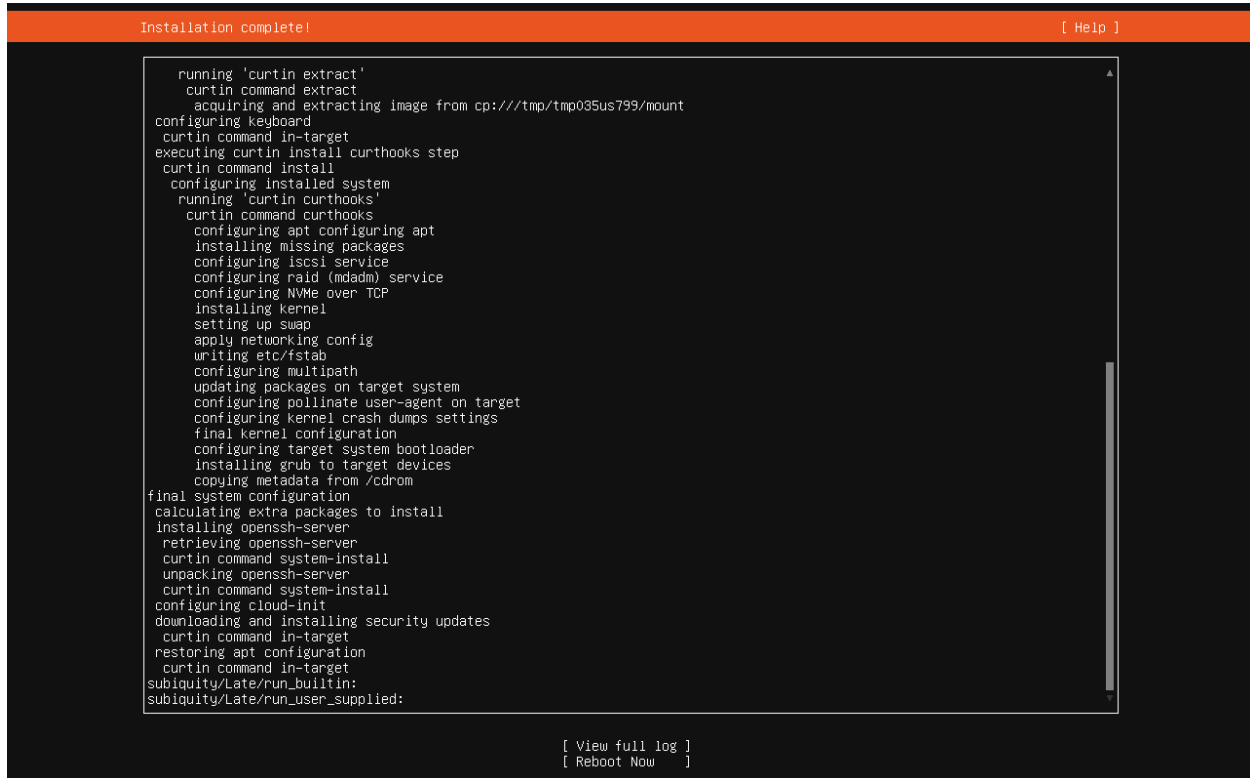
[Done]

5. Enable SSH

Select Install OpenSSH server. Skip extra packages unless instructed.

6. Complete installation

Wait for completion, reboot, and remove the ISO if prompted.



```
Installation complete! [ Help ]

running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmp035us799/mount
configuring keyboard
  curtin command in-target
executing curtin install curthooks step
  curtin command install
    configuring installed system
      running 'curtin curthooks'
        curtin command curthooks
          configuring apt
            configuring apt
              installing missing packages
                configuring iscsi service
                configuring raid (mdadm) service
                configuring NVMe over TCP
                installing kernel
                setting up swap
                apply networking config
                writing etc/fstab
                configuring multipath
                updating packages on target system
                configuring pollinate user-agent on target
                configuring kernel crash dumps settings
                final kernel configuration
                configuring target system bootloader
                installing grub to target devices
                copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
  retrieving openssh-server
  curtin command system-install
unpacking openssh-server
  curtin command system-install
configuring cloud-init
downloading and installing security updates
  curtin command in-target
restoring apt configuration
  curtin command in-target
subiquity/late/run_builtin:
subiquity/late/run_user_supplied:

[ View full log ]
[ Reboot Now ]
```


4. Initial Configuration and Verification

Log in with the non-root administrative user. Run each command and capture the result. Do not expose passwords, private keys, tokens, or unrelated personal information.

```
hostname
```

```
Ubuntu 26.04 LTS server01 tty1
server01 login: rizannealay
Password:
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-29-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Aug 15 11:11:38 AM UTC 2026

System load:          0.32
Usage of /:            37.6% of 18.53GB
Memory usage:         8%
Swap usage:           0%
Processes:            125
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe3f:33f1

Expanded Security Maintenance for Applications is not enabled.

48 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

rizannealay@server01:~$ hostname
server01
rizannealay@server01:~$ _
```

ip addr

```
* Management: https://landscape.canonical.com
* Support:    https://ubuntu.com/pro

System Information as of Sat Aug 15 11:11:38 AM UTC 2026

System load:          0.32
Usage of /:            37.6% of 10.53GB
Memory usage:         0%
Swap usage:           0%
Processes:            125
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe3f:33f1

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applicable law.

rizannealay@server01:~$ hostname
server01
rizannealay@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:00:27:3f:33:f1 brd ff:ff:ff:ff:ff:ff
    altname enx0000273f33f1
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86219sec preferred_lft 86219sec
    inet6 fd17:625c:f037:2:a00:27ff:fe3f:33f1/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86220sec preferred_lft 14220sec
    inet6 fe80::a00:27ff:fe3f:33f1/64 scope link
        valid_lft forever preferred_lft forever
rizannealay@server01:~$
```

ping -c 4 google.com

```
Expanded Security Maintenance for Applications is not enabled.

48 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

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applicable law.

rizannealay@server01:~$ hostname
server01
rizannealay@server01:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:3f:33:f1 brd ff:ff:ff:ff:ff:ff
    altname enx0800273f33f1
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86219sec preferred_lft 86219sec
    inet6 fd17:625c:f037:2:a00:27ff:fe3f:33f1/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86220sec preferred_lft 14220sec
    inet6 fe80::a00:27ff:fe3f:33f1/64 scope link proto kernel ll
        valid_lft forever preferred_lft forever
rizannealay@server01:~$ whoiam
whoiam: command not found
rizannealay@server01:~$ whoami
rizannealay
rizannealay@server01:~$ ping -c 4 google.com
PING google.com (172.217.26.238) 56(84) bytes of data:
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=1 ttl=64 time=10.8 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=2 ttl=64 time=9.38 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=3 ttl=64 time=9.69 ms
64 bytes from nrt12s51-in-f14.1e100.net (172.217.26.238): icmp_seq=4 ttl=64 time=8.78 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3053ms
rtt min/avg/max/mdev = 8.778/9.671/10.846/0.752 ms
rizannealay@server01:~$
```

```
sudo apt update
sudo apt upgrade -y
```

```
server01 login: rizannealay
Password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.8.0-137-generic x86_64)
```

```
* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro
```

System information as of Sat Aug 15 01:58:24 PM UTC 2026

```
System load:          0.68
Usage of /:            34.7% of 18.53GB
Memory usage:         5%
Swap usage:           0%
Processes:            111
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fee7:136c
```

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See <https://ubuntu.com/esm> or run: `sudo pro status`

```
rizannealay@server01:~$ sudo apt update
[sudo] password for rizannealay:
Hit:1 http://ph.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://ph.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://ph.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
5 packages can be upgraded. Run 'apt list --upgradable' to see them.
rizannealay@server01:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following packages were automatically installed and are no longer required:
  libfwupd2 libgusb2
Use 'sudo apt autoremove' to remove them.
The following upgrades have been deferred due to phasing:
  krb5-locales libgssapi-krb5-2 libk5crypto3 libkrb5-3 libkrb5support0
0 upgraded, 0 newly installed, 0 to remove and 5 not upgraded.
rizannealay@server01:~$
```

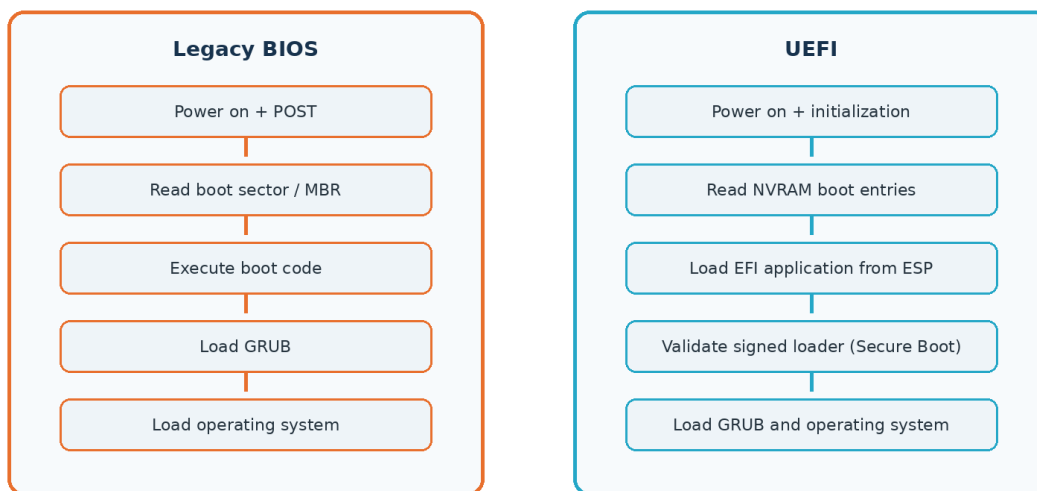
```
systemctl status ssh
```

```
rizannealay@server01:~$ systemctl status ssh
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sat 2026-08-15 14:05:56 UTC; 2min 18s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 1340 (sshd)
      Tasks: 1 (limit: 4600)
     Memory: 1.2M (peak: 1.3M)
        CPU: 37ms
     CGroup: /system.slice/ssh.service
            └─1340 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 15 14:05:56 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 15 14:05:56 server01 sshd[1340]: Server listening on 0.0.0.0 port 22.
Aug 15 14:05:56 server01 sshd[1340]: Server listening on :: port 22.
Aug 15 14:05:56 server01 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
rizannealay@server01:~$
```

5. BIOS vs UEFI

Firmware Boot Paths: Legacy BIOS and UEFI

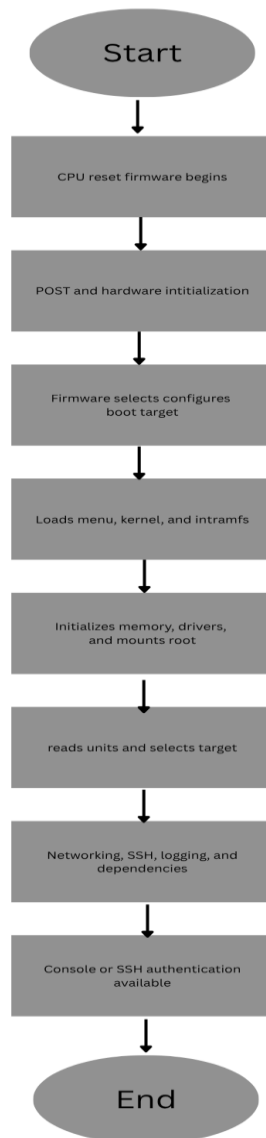


Firmware boot paths: Legacy BIOS and UEFI

BIOS executes legacy boot code from a disk boot sector, while UEFI uses standardized firmware interfaces, NVRAM boot entries, and EFI applications stored on an EFI System Partition. UEFI is preferred for most modern deployments because it supports GPT, flexible boot management, modern hardware initialization, and Secure Boot.

6. Ubuntu Boot Process

Ubuntu Server Boot Process



Ubuntu Server boot process

Firmware selects the boot target and starts GRUB. GRUB loads the Linux kernel and initramfs. The kernel initializes hardware and mounts the root file system, then starts systemd as PID 1. systemd activates services until local or remote login is available.

7. Windows Server Installation Summary

A separate VM using Windows Server Evaluation. Select the required edition and installation type, install to the approved virtual disk, assign a compliant Administrator password, sign in, and rename the computer.

Field	Recorded value
Edition	Windows Server 2025
VM name	Oracle Virtual Box
Computer name	Windows-Server-Week03
RAM / CPU / disk	4096 mb / 2 / 40
Network	Intel PRO / 1000 MT Desktop (NAT)

Windows Server installation

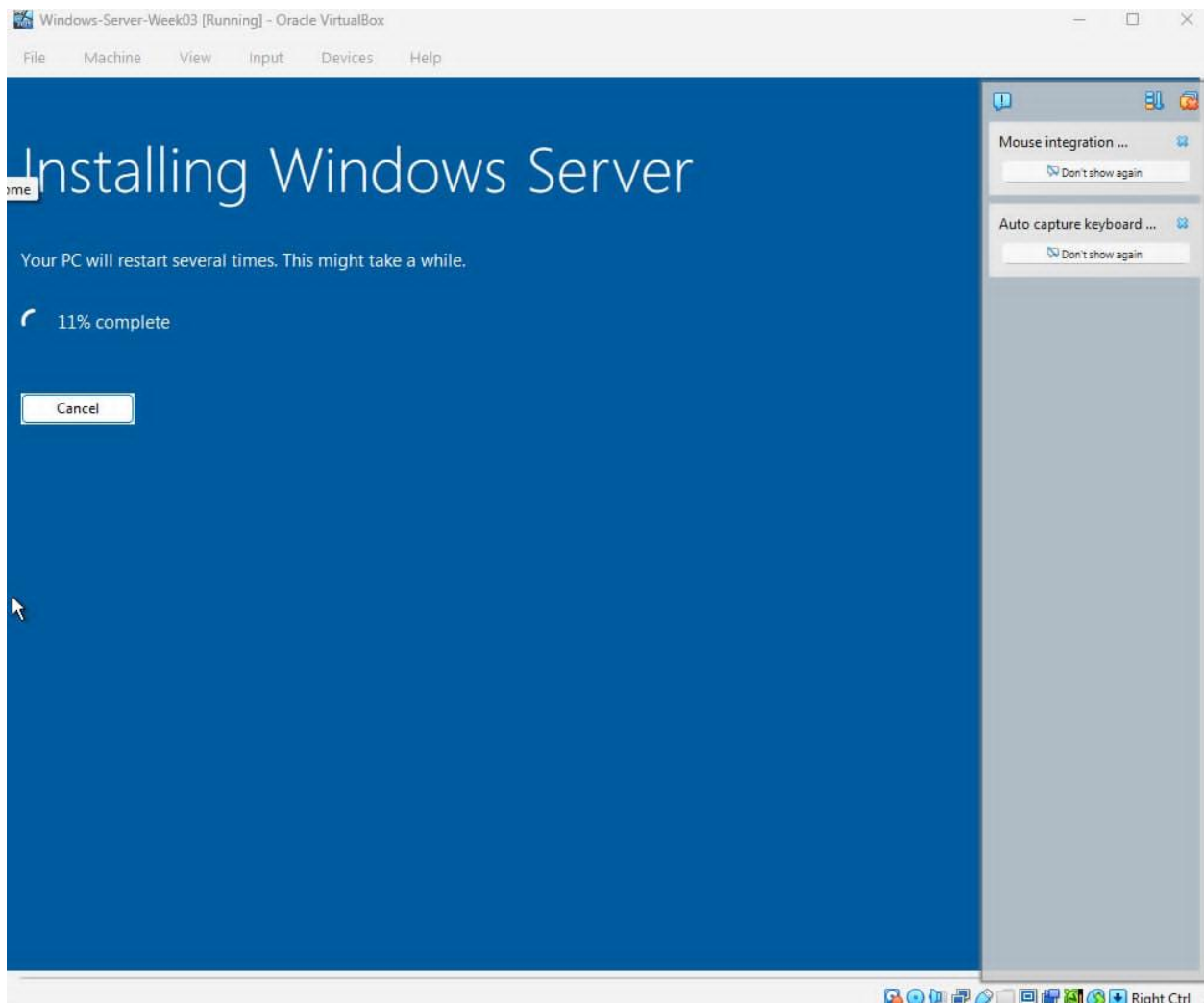
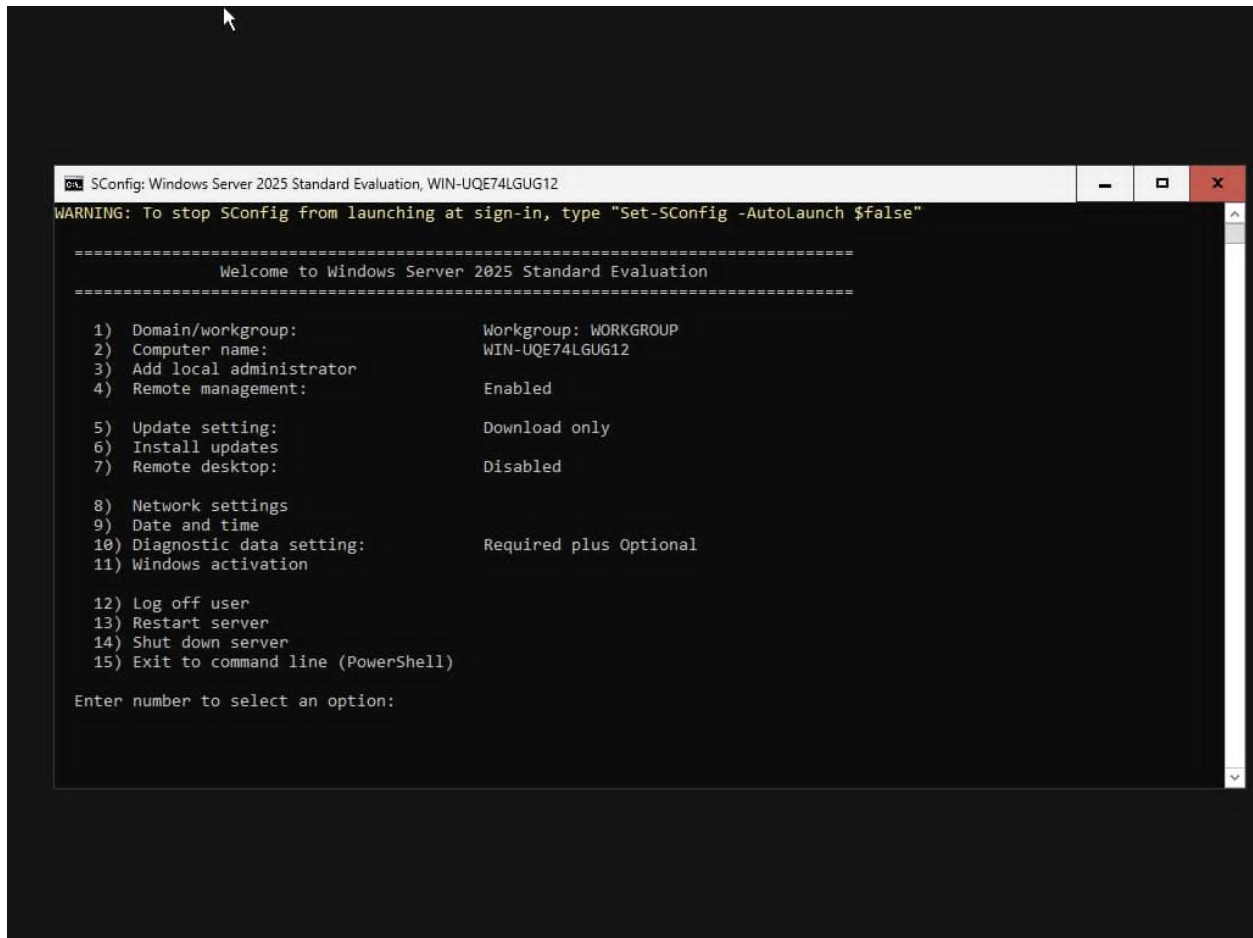


Figure: Successful Windows Server login



8. Enterprise OS Comparison

Area	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Commercial; evaluation available	Open source; optional paid support	Open-source community enterprise Linux
Interface	GUI and Server Core	CLI focused; GUI optional	CLI focused; GUI optional
Packages	Windows Update and roles/features	APT / DEB	DNF / RPM
Security	Defender, firewall, AD controls	AppArmor, firewall, security updates	SELinux, firewalld, enterprise lifecycle
Typical use	AD and Microsoft workloads	Cloud, web, containers, DevOps	RHEL-compatible enterprise services
Advantage	Integrated Microsoft ecosystem	Broad cloud and community support	Stable RHEL compatibility
Disadvantage	Licensing cost	CLI learning curve	Conservative package versions

Discussion

Windows Server is a strong fit where Active Directory and Microsoft application integration are central. Ubuntu Server suits cloud-native, web, container, and development workloads. Rocky Linux provides a stable RHEL-compatible platform. Selection should consider application support, administrator skills, lifecycle, security policy, and total cost.

9. Problems, Solutions, and Lessons

Problem encountered	Solution implemented	Verified result
After completing the Ubuntu Server installation, the virtual machine started the installation wizard again instead of loading the installed operating system.	I powered off the virtual machine, removed the Ubuntu Server ISO from the virtual optical drive, and changed the boot order so the virtual hard disk would start first.	The virtual machine successfully booted from the virtual hard disk and displayed the Ubuntu Server login prompt. I logged in using the administrative account I created.

Lessons learned

Throughout this project, I learned that installing a server operating system involves more than simply following the instructions shown by an installation wizard. Proper server deployment requires careful planning, correct configuration, testing, security awareness, and complete documentation. While creating the Ubuntu Server virtual machine, I understood the importance of assigning appropriate resources such as memory, virtual processors, storage capacity, network mode, and installation media. I also learned that selecting the correct virtual disk is essential because using the guided entire-disk option can erase all information on the selected storage device. During the installation, I configured the hostname as `server01`, created a non-root administrative account, used DHCP for network configuration, and enabled OpenSSH Server for remote administration. These tasks helped me understand why server identity, account security, networking, and remote-access services must be properly configured before a server is deployed. The verification activities taught me not to assume that an installation is successful based only on reaching the login screen. I used commands such as `hostname`, `ip addr`, `ping -c 4 google.com`, `sudo apt update`, `sudo apt upgrade -y`, and `systemctl status ssh` to confirm that the server was correctly named, connected to the network, able to access the internet, updated, and ready for SSH connections.

I also learned how screenshots provide important evidence that each requirement has been completed and verified. Studying BIOS and UEFI helped me understand what happens before the operating system starts. BIOS is an older firmware system commonly associated with MBR, while UEFI supports modern features such as GPT, larger storage devices, flexible boot management, and Secure Boot. Creating the Ubuntu boot-process flowchart helped me understand the sequence from powering on the computer to firmware initialization, boot device detection, GRUB, the Linux kernel, `systemd`, service startup, and the login prompt. Comparing Windows Server, Ubuntu Server, and Rocky Linux also showed me that each operating system has different strengths, licensing models, package-management systems, security tools, and enterprise uses. Windows Server is suitable for Microsoft-centered environments, Ubuntu Server is widely used for cloud, web, and development workloads, and Rocky Linux provides a stable RHEL-compatible environment. Overall, this project improved my confidence in server installation, command-line administration, troubleshooting, verification, and technical documentation. Most importantly, I learned that professional system administrators must make careful decisions, protect sensitive information, verify every important configuration, and create clear documentation that another administrator can reliably follow.

10. References

Official Ubuntu, UEFI, Microsoft, Rocky Linux, GNU GRUB, `systemd`, and virtualization documentation were consulted. Accessed 15 August 2026.